```c
#include <stdio.h>
#include <math.h>

#define READING_COUNT 5

double calculateAverage(double readings[], int count) {
    double total = 0;
    for (int i = 0; i < count; i++) total += readings[i];
    return total / count;
}

void displayReadings(double readings[], int count) {
    printf("Temperature Readings:\n");
    for (int i = 0; i < count; i++) printf("%.1f F\n", readings[i]);
}

int isOutOfRange(double temperature) {
    //return (temperature > 75 || temperature < 65) ? 1 : 0;
    return (temperature > 75 || temperature < 68) ? 1 : 0;    //change range
}

void checkReadings(double readings[], int count) {
    printf("\nReadings requiring attention:\n");
    for (int i = 0; i < count; i++) {
        if (isOutOfRange(readings[i])) printf("%.1f F\n", readings[i]);
    }
}

int testsRun = 0;
int testsPassed = 0;

void recordTest(const char *testName, int passed)
{
    testsRun++;
    if (passed) {
        testsPassed++;
        printf("PASS: %s\n", testName);
    } else {
        printf("FAIL: %s\n", testName);
    }
}

/* Section 3 - Initial Test Suite */

/* T1: 75 F is within the acceptable range. */
void test_T1_normalTemperature(void) {
    double temperature = 75.0;
    recordTest("T1 - 75 F is within the acceptable range",
               isOutOfRange(temperature) == 0);
}

/* T2: 64 F is below the minimum and should require attention. */
void test_T2_belowMinimum(void) {
    double temperature = 64.0;
    recordTest("T2 - 64 F requires attention",
               isOutOfRange(temperature) == 1);
}

/* T3: 78.5 F is above the maximum and should require attention. */
```

```
void test_T3_aboveMaximum(void) {
    double temperature = 76;
    recordTest("T3 - 76 F requires attention",
               isOutOfRange(temperature) == 1);
}

void runInitialTests(void) {
    printf("\n=======================================\n");
    printf("CSE 1106 INITIAL TEST SUITE\n");
    printf("=======================================\n\n");

    test_T1_normalTemperature();
    test_T2_belowMinimum();
    test_T3_aboveMaximum();
}

/* Add your tests T4 - T8 here. */
/*Tests should follow the same flow and invoke the recordTest method*/
/* T4: One value below the minimum. */
void test_T4(void) {
    double temperature = 52.0;
    recordTest("T2 - 52 F requires attention",
               isOutOfRange(temperature) == 1);
}

/* T5: The minimum */
void test_T5(void) {
    double temperature = 68.0;
    recordTest("T1 - 68 F is within the acceptable range",
               isOutOfRange(temperature) == 0);

}
/* T6: One value inside the range. */
void test_T6(void) {
    double temperature = 72.0;
    recordTest("T1 - 72 F is within the acceptable range",
               isOutOfRange(temperature) == 0);

}
/* T7: The maximum. */
void test_T7(void) {
    double temperature = 75.0;
    recordTest("T1 - 75 F is within the acceptable range",
               isOutOfRange(temperature) == 0);

}
/* T8: One value above the maximum. */
void test_T8(void) {
    double temperature = 80;
    recordTest("T3 - 80 F requires attention",
               isOutOfRange(temperature) == 1);

}

/* Finish the method "runTestsNewRequirement" below that invokes the minimal number
of tests from T1-T8 needed to ensure (1) the new template change is correctly coded
and (2) nothing else has changed. */
void runTestsNewRequirement(void) {
    printf("\n=======================================\n");
```

```
    printf("CSE 1106 NEW REQUIREMENT TEST SUITE\n");
    printf("=====================================\n\n");

    //TODO invoke relevant tests
    test_T4();
    test_T5();
    test_T6();
    test_T7();
    test_T8();
}
int main(void) {
    double readings[READING_COUNT] = {68.0, 69.0, 70.0, 71.0, 100.0};

    printf("=====================================\n");
    printf("LEGACY CLASSROOM TEMPERATURE MONITOR\n");
    printf("=====================================\n\n");

    /* First observe the existing program. */
    displayReadings(readings, READING_COUNT);
    printf("\nAverage temperature: %.1f F\n",
           calculateAverage(readings, READING_COUNT));
    checkReadings(readings, READING_COUNT);

    /* Then run the initial test suite. */
    runInitialTests();

    printf("\n=====================================\n");
    printf("TEST SUMMARY\n");
    printf("=====================================\n");
    printf("Tests passed: %d/%d\n", testsPassed, testsRun);

      testsRun = 0;
      testsPassed = 0;

      /* Uncomment the below code to run the new requirement test suite for section
6.  */

    runTestsNewRequirement();

    printf("\n=====================================\n");
    printf("TEST SUMMARY\n");
    printf("=====================================\n");
    printf("Tests passed: %d/%d\n", testsPassed, testsRun);

    return 0;
}
```